The PTB Simulation Report

User: Campbell

User Email: mc53@email.sc.edu

Total Score: 97/100

Duration Score: 33/34

Cost Score: 31/33

Scope Score: 33/33

Scenario: Transformation-B

Created at: Thursday, November 16, 2023 4:58:08 PM

Last simulation run result: Scenario finished

Simulation Time: 33

Target Date: 32

Target Cost: 129000

Simulation Cash: 26908

Total Cost: 138093

Value: 100

Estimated Cost: 138093

Estimated Duration: 33

Morale: Average

Number of Deliverables Used: 3

Risks:

Name	Type	Affected Task	Probability	Mitigation Cost	Effect	Is Mitigated
Technological Failure	Performance		30	4000	20	False
Integration Issue	Task Cost	Connection to Grid	50	1000	200	False

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Question: What did you learn from running the simulation?

Answer: I learned how much morale helps with duration

Question: In case you worked on this project in a team, how did you distribute the management of the project between the team members?

Answer: We worked together to find the best way to do the assigmnent

Question: How did you manage the risk in the project?

Answer: By focusing on it first and trying to get it done in a effective amount of time

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Question: What did you find most useful or enjoyable?

Answer: Being able to manage everything to test for the best outcome

Question: Please enter any comments and recommendations:

Answer: .

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Scenario Information:

Project Name	Number of Tasks	Target Date	Initial Cash	Bonus per Day	Penalty per Day
Main Project	12	32	165000	1000	5000

Resource Information:

Name	Available Units	Available Units in Lead Time	Cost/ Day	Idle Cost/ Day	Firing Cost	Hiring Cost	Show Up Probability (%)	Firing Lead Time	Hiring Lead Time	Minimum Units	Maximum Units
Workshopworkers	3	3	450	100	30	30	95	1	1	1	7
LV workers	5	5	400	100	30	30	93	1	1	1	12
HV workers	5	5	400	100	30	30	90	1	1	1	13

Quality Requirements:

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Name	Formula	Importance	Minimum Value	Desired Value	Maximum Value	Best Mode	Evaluation
Reliability	[TR]*[RHTS]*[RLTS]*[RHTL]*[RLTL]*200	1	0	70	100	Maximum	61.56

Task Information:

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Construction	1	0	12	0	Finished	Subcontractor A

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshop workers Needed	LV workers Needed	HV workers Needed
Subcontractor A	10	11	12	0	0	0	0

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Manufacture HV Switchboards	1	1	4	0	Finished	Subcontractor C

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshop workers Needed	LV workers Needed	HV workers Needed
Subcontractor A	1	3	5	3500	5	0	0
Subcontractor B	3	4	5	3500	4	0	0
Subcontractor C	3	4	6	1000	2	0	0

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Manufacture LV Switchboards	1	1	9	0	Finished	Subcontractor C

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshop workers Needed	LV workers Needed	HV workers Needed
Subcontractor A	1	3	4	2500	5	0	0
Subcontractor B	4	6	9	3200	4	0	0
Subcontractor C	4	6	9	1000	2	0	0

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Install HV Transformer	1	12	19	0	Finished	Subcontractor B

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshopworkers Needed	LV workers Needed	HV workers Needed	Transformer Reliability (TR) (TR)
Subcontractor A	4	5	11	8500	0	0	5	0.8
Subcontractor B	6	7	8	4000	0	0	2	0.5

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Install HV Switchboards	1	12	16	0	Finished	Subcontractor B

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshopworkers Needed	LV workers Needed	HV workers Needed	HV Switchboard Reliabiliy (RHTS) (RHTS)
Subcontractor A	1	3	5	2500	0	0	3	0.9
Subcontractor B	2	4	5	1100	1	0	2	0.9
Subcontractor C	3	5	10	800	0	0	2	0.8

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Install LV Switchboards	1	13	15	0	Finished	Subcontractor A

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshopworkers Needed	LV workers Needed	HV workers Needed	LV Switchboard Reliability (RLTS) (RLTS)
Subcontractor A	1	3	7	7700	2	0	5	0.8
Subcontractor B	2	4	5	7500	2	0	4	0.7
Subcontractor C	3	5	8	1500	0	0	2	0.3

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Install HV Lines	1	19	29	0	Finished	Subcontractor B

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshopworkers Needed	LV workers Needed	HV workers Needed	HV Lines Reliability (RHTL) (RHTL)
Subcontractor A	5	7	9	5500	0	0	5	0.9
Subcontractor B	8	9	10	3000	0	0	3	0.95

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Install LV lines	1	19	27	0	Finished	Subcontractor B

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshopworkers Needed	LV workers Needed	HV workers Needed	LV Lines Relaibility (RLTL) (RLTL)
Subcontractor A	2	4	6	2000	0	3	0	0.6
Subcontractor B	5	7	9	3000	0	3	0	0.9

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Connection to Grid	1	29	31	0	Finished	Subcontractor A

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshop workers Needed	LV workers Needed	HV workers Needed
Subcontractor A	1	2	3	2200	1	2	1
Subcontractor B	2	3	4	1300	1	1	1

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Install Generator	0	12	14	0	Finished	Subcontractor B

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshop workers Needed	LV workers Needed	HV workers Needed
Subcontractor A	1	2	6	7200	0	3	0
Subcontractor B	1	3	5	6100	0	2	0

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Connect Generator to Grid	1	15	18	0	Finished	Subcontractor B

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshop workers Needed	LV workers Needed	HV workers Needed
Subcontractor A	3	5	9	1000	0	5	0
Subcontractor B	2	3	5	5500	0	3	0

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Project Name	Task Name	Cost of Split	Actual Start	Actual Finish	Income	Current Status	Selected Mode
Main Project	Turn-on and Delivery	1	31	33	1	Finished	Subcontractor A

Mode Name	Optimistic Duration	Most Likely Duration	Pessimistic Duration	Fixed Cost	Workshop workers Needed	LV workers Needed	HV workers Needed
Subcontractor A	1	2	3	2300	1	2	1
Subcontractor B	2	3	4	1400	1	1	1